

Table 1: Summary of material properties and constituent assumptions¹

Component	Material	Primary function	Constituent Model	Required Properties
Main frame	Carbon fibre (CFRP)	Structural integrity & weight minimization	Linear-elastic orthotropic carbon/epoxy lamina, assembled as a symmetric quasi-isotropic laminate. Material coordinate system follows the principal fibre direction of the frame.	$E_1 = 135 \text{ GPa}$; $E_2 = E_3 = 10 \text{ GPa}$; $V_{12} = 0.30$; $V_{13} = 0.3$; $V_{23} = 0.4$; $G_{12} = 5.0$; $G_{13} = 5.0$; $G_{23} = 3.8$ $P = 1600 \text{ kgm}^{-2}$; $X_t = 1500 \text{ Mpa}$; $X_c = 1200 \text{ Mpa}$; $Y_t = 50 \text{ Mpa}$; $Y_c = 200 \text{ Mpa}$; $S_{12} = 70 \text{ Mpa}$ Laminate = $[0/ + 45/ - 45/90]_{st}$ with 0.25mm ply thickness; frame thickness = 5mm uniform Frame wt. original = 5.1kg, Optimized = 4.1kg
Active Suspension Hinge (ASH)	CFRP/ Ti-6Al-4V	Impact force attenuation	Isotropic linear-elastic Titanium alloy solid; non-linear contact at hinge interfaces if relative articulation is represented.	$E = 114 \text{ GPa}$; $V = 0.34$; $\rho = 4430 \text{ kgm}^{-3}$; Yield strength $\sigma_y = 880 \text{ Mpa}$ Ultimate tensile strength $\sigma_u = 950 \text{ Mpa}$ From 2.5mm to 6mm.
Spring Damper	Ti-6Al-4V	Impact force attenuation	Isotropic Titanium structural solid combined with a spring damper connector model	$E = 114 \text{ GPa}$; $V = 0.34$; $\rho = 4430 \text{ kgm}^{-3}$; $\sigma_y = 880 \text{ Mpa}$; C ; $K =$
Telescopic support rods	Grade 5 Ti / Al Alloy	Load by-pass to the ground	Telescoping structural members with contact between the internal & external tube sections.	Ti-6Al-4V: $E = 114 \text{ GPa}$; $V = 0.34$; $\rho = 4430 \text{ kgm}^{-3}$; $\sigma_y = 880 \text{ Mpa}$; Al alloy (7075-T6): $E = 71.7 \text{ GPa}$; $V = 0.33$; $\rho = 2810 \text{ kgm}^{-3}$; $\sigma_y = 503 \text{ Mpa}$;
Omni-Wheels (OMDW)	CFRP rim/wheel body	Multi-axis mobility on narrow paths	Linear, elastic orthotropic carbon/epoxy laminate. Local material axes follows the principal fibre directions around the wheel structure.	$E_1 = 135 \text{ GPa}$; $E_2 = 10 \text{ GPa}$; $E_3 = 10 \text{ GPa}$; $V_{12} = 0.30$; $V_{13} = 0.3$; $V_{23} = 0.4$; $G_{12} = 5.0$; $G_{13} = 5.0$; $G_{23} = 3.8$ $P = 1600 \text{ kgm}^{-2}$; $X_t = 1500 \text{ Mpa}$; $X_c = 1200 \text{ Mpa}$; $Y_t = 50 \text{ Mpa}$; $Y_c = 200 \text{ Mpa}$; $S_{12} = 70 \text{ Mpa}$ Laminate = $[0/ + 45/ - 45/90]_{st}$ with 0.25mm ply thickness
	Titanium Hub & Axle		Isotropic, linear-elastic structural solid	$E = 114 \text{ GPa}$; $V = 0.34$; $\rho = 4430 \text{ kgm}^{-3}$; $\sigma_y = 880 \text{ Mpa}$; $\sigma_u = 950 \text{ Mpa}$
	Polyurethane roller tread		Isotropic, hyper-elastic or simplified linear elastic compliant layer.	Linear elastic approx. $E = 25 \text{ Mpa}$; $V = 0.49$; $\rho = 1100 \text{ kgm}^{-3}$ Nominal thread harness: Shore A 85 – 95
	Wheel ground interface		Nonlinear frictional contact or compression only support simplified static load transfer.	Dry rock friction coefficient (μ) = 0.6; Sensitivity range = $0.4 \leq \mu \leq 0.70$
	Rolling resistance		Equivalent resisting moment or force opposing wheel rotation & translation	Coefficient of resistance: - For firm, dry terrain, $C_{RR} = 0.03$ - Rough, loose or rocky terrain, $C_{RR} = 0.05 - 0.10$ Resistive force (F_{RR}) = $C_{RR}N$
Ergonomic mesh harness	Antimicrobial Nylon mesh/ Velcro	User comfort & heat dissipation	Flexible orthotropic fabric or simplified compliant load-transfer interface.	$G_{warp} = 0.5 \text{ GPa}$; $E_{weft} = 0.35 \text{ GPa}$; $V = 0.39$; $\rho = 1150 \text{ kgm}^{-3}$; Tensile strength = 70 Mpa; nominal thickness = 2.0 mm

Source: FreeCAD/Authors

¹ The structural analysis assumes linear elastic behaviour within the range of the applied loads. No claim of progressive composite damage, delamination or post-failure behaviour is made in the present analysis.

Table 4: Anthropometric and biomechanical properties of the virtual human model

Parameter	Value	Source/Justification
Sex	Female	Study design
Anthropometric percentile	50 th	Study design
Body mass	75kg	Prescribed simulation condition
Stature	1.65 m	Representative modelling estimate
Head & neck mass	5.01 kg	6.68% of body mass [47]
Trunk mass	31.93 kg	42.57 of body mass [47]
Upper arm mass	1.91 kg each	2.55% per segment [47]
Forearm mass	1.04 kg each	1.38% per segment [47]
Hand mass	0.42 kg each	0.56% per segment [47]
Thigh mass	11.09 kg each	14.78% per segment [47]
Shank mass	3.61 kg each	4.81% per segment [47]
Foot mass	0.97 kg each	1.29% per segment [47]
Head & neck segment length	≈ 0.21 m	Representative anthropometric estimate
Trunk length	≈ 0.52 m	Representative anthropometric estimate
Upper arm length	≈ 0.305 m each	Representative anthropometric estimate
Forearm length	≈ 0.240 m each	Representative anthropometric estimate
Hand length	≈ 0.175 m each	Representative anthropometric estimate
Thigh length	≈ 0.405 m each	Representative anthropometric estimate
Shank length	≈ 0.400 m each	Representative anthropometric estimate
Foot length	≈ 0.250 m each	Representative anthropometric estimate
Cervical/head joint	3 rotational DOF	Head-trunk articulation
L5/S1 joint	3 rotational DOF	Trunk-pelvis articulation & spinal-load assessment
Hip joints	3 rotational DOF each	Spherical hip representation
Knee joints	1 primary rotational DOF each	Each Flexion/extension
Ankle joints	1 primary rotational DOF each	Each Dorsiflexion/plantar flexion
Shoulder joints	3 rotational DOF each	Each spherical shoulder representation
Elbow joints	1 primary rotational DOF each	Each Flexion/extension
Wrists joints	2 rotational DOF each	Each Flexion/extension & radial/inner deviation
Walking speed	1.2 m/s	Simulation condition
Terrain inclination	20 deg.	Simulation condition
Payload	40kg (≈ 400 N)	Design-standard condition
Load carriage distance	4.3 Km	Field data for LCA operational estimation
Trips per day	2.77	Field data for LCA operational estimation
Assessment period	5 years	Field data for LCA operational estimation

Note: segment mass and lengths were computed using the female segment masses fractions reported by de leva [47]. This provides adjusted anthropometric segment parameters referenced to commonly as anatomical joint centres.

For each body segment; $M_i = f_i M$

Where

M_i = Mass of segment I,

f_i = female segment mass fraction;

M = total body mass (75 kg)

Table 5: Summary of principal MBD connections

Human-Device Component	MBD Connections	Mechanical function
Shoulder/Lifter straps - human/frame	Fixed attachment	Transfers load between the harness & frame
Lumbar/Hip belt – human/frame	Fixed attachment	Maintains frame-body alignment & transfers interface forces
CFRP frame	Rigid body	Provides structural backbone of the device
CFRP frame – Active suspension hinge	Revolute joint	Permits controlled angular motion about the hinge axis
Hinge to Telescopic support assembly	Rigid/Revolute connection	Transfers suspension motion & force to support system
Telescopic rod	Translational/Prismatic joint	Permits axial extension & compression
Spring damper - Telescopic rod	Compliant force element	Represents elastic & damping response
Telescopic rod - wheel assembly	Rigid/Appropriate joint	Transfers support forces to wheel
Wheel – Wheel axle	Revolute joint	Allows wheel rotation
Wheel - Terrain	Contact/Friction interaction	Transfers load to the ground & governs traction/slip

Table 6

Payload Coordinates relative to Stance Foot contact - A					
Link (i)	Segment Description	Length L_i (m)	Angle (ϕ_i) Degrees	$\Delta X_i = L_i \cos \phi_i$ (m)	$\Delta Y = L_i \sin \phi_i$ (m)
1	Shank	0.400	75.0	+0.1035	+0.3864
2	Thigh	0.405	65.0	+0.1712	+0.3671
3	Trunk (Hip – Mount)	0.520	80.0	+0.0903	+0.5121
4	Frame Stand-off)	0.240	170.0	-0.2364	+0.0417
5	Trailing Strut (L_s)	0.900	215.0	-0.7372	-0.5162
Payload position relative to OMDW contact					
Horizontal distance to rear wheel (X_L) = -0.6086 m (60.86 cm)					
Vertical height above wheel (Y_L) = +0.7911 m (79.11 cm)					
Resultant payload vector magnitude = PL = 0.9981 m					
Summary Virtual Human Body Mass inputs - B					
Segment	Mass per Unit (Kg)		Quantity	Total Sub-mass (Kg)	
Head/Neck	5.01		1	5.01	
Trunk	31.93		1	31.93	
Upper arms	1.91		2	3.92	
Forearms	1.04		2	2.08	
Hand	0.42		2	0.84	
Thigh	11.09		2	22.18	
Shank	3.61		2	7.22	
Foot	0.97		2	1.94	
Total Virtual Human Body Mass	-		-	75.12	
Payload weight (W_P) = 40 kg * 9.81 m/s ² = 400 N					
Human Body Weight (W_{body}) = 75 * 9.81) = 735.9 N					
Total Combined Gravity Load (W_{Total}) = 1135.75 N (1128.34 – 1135.75)					
	Summary of L5/S1 and Hip dynamic joint burden - C				
Segment	Mass per Unit (Kg) (refer Annexure A)		Quantity	Total Sub-mass (Kg)	
Head/Neck	5.01		1	5.01	
Trunk	31.93		1	31.93	
Upper arms	1.91		2	3.92	
Forearms	1.04		2	2.08	
Hand	0.42		2	0.84	
Total Upper body mass (M_{Upper})	-		-	43.68	

Net compressive force across L5/S1 Joint with LoCaDe-Exo bypass ($F_{L5/S1}$) = 548.5 N				
Total compressive force ($F_{Unassisted}$)				
Because there is no bypass mechanism in the unassisted condition, 100% of the payload weight ($W_{payload}$) is borne directly by the torso/spine				
Therefore, $F_{Unassisted}$) = $W_{Upper\ body} + W_{payload} = 428.50\text{ N} + 400\text{ N} = 828.50\text{ N}$				
Summary of force transmission model outputs - D				
Variable	Governing formulae	Static value	Peak transient value	Unit
Normal gravitational term	$F_{Load} * \text{Cos}\theta$	375.88	568.38	N
Active Suspension Hinge resistance (F_{damp})	$C\dot{x} + Kx$	106.00	$87 + 61 = 148.50$	N
Ground normal force (F_{ground})	$F_{load} \text{Cos}\theta - F_{damp}$	269.28	418.88	N
Ground shunt percentage (%)	$(F_{ground}/F_{load})*100$	71.68	73.8	%
$F_{Load} = 40\text{ kg} * 9.9812 = 392.4\text{ N}$ (or 400N nominal) $F_{Load} * \text{Cos } \theta = 392.4 \text{ Cos } 20^0 = 368.7\text{ N}$				
Global Force Equilibrium Formulation (GFE) - E				
Payload weight (W_p) = $40\text{ kg} * 9.81\text{ m/s}^2 = 400\text{ N}$ Total Combined Gravity Load (W_{Total}) = 1135.75 N Ground Shunt percentage = 71.68%				
Ground Shunt proportion	N_{Shunt}	286.72 N		
Physiological retention proportion	$N_{Ret.}$	113.48 N		
Ground Reaction Force at Omni-wheel (OMDW) Contact - F				
Ground vertical GRF	$F_y, wheel$	286.52		N
Terrain normal force	F_n	269.24		N
Terrain-Longitudinal Friction/Downslope force	F_{II}	97.98		N
Terrain margin verification	$U_{req.} = F_{II} / F_n$ $= \tan (20^0)$	0.364		
Payload offset distance	$d_{p,H}$	0.22		M
Effective moment arm	$L_{Movement}$	0.733		M
F_{strut}	$W_p * d_{p,H} / L_{Movement}$	120.5 nominal		N
F_{human}	$F_{human} = F_{strut}$	≈ 120		N
F_{ground}	$W_p - F_{strut}$	≈ 280		N
$U_{req.} \approx 0.364 \leq U_{dry,rock} (0.60)$ Safety Factor against wheel slip (SF) = $U_{terrain}/U_{req.} = 0.60/0.360 = 1.65$ (No Slip) $\sum M_H = 0 \Rightarrow W_p * d_{p,H} - F_{Strut} * L_{Movement} = 0 \Rightarrow 88 - 120 * 0.733 = 0$				
Where,				
$d_{p,H}$ = payload centre of mass offset behind Hip pivot $L_{Movement}$ = perpendicular lever arm of the trailing strut				

Table 8:

Field Data from Dunda Block, Uttarkashi district, Uttarakhand state				
Variable	Distance covered during load carriage?	How many trips do you need to make to complete a task?	On average, how much weight do you carry in a single load? (kg)	How many days do you carry loads in a month?
Mean	4.3836	2.77	≈ 40	26
SD	3.99	0.8604	12.55	
Pandolf-Santee Model				
D_{month}	316.3 Km/month			
$E_{Life,i}$	18,980 Km			
Human mass (W)	75 Kg			
Payload (L)	40 Kg			
Walking speed	1.2m/s			
Inclination	20 deg.			
Grade (G)	36.40%			
EF _{food}	= 0.20 Kg CO ₂ e/MJ			

Table 12: Modal analysis parameters

• Cadence (step frequency) = $F = V/L_{step}$ • Typical walking level-ground step length = 0.55 – 0.7 m • Adjustment for 20 deg. Incline 0.47 – 0.6 m • Natural frequency gait = $F_1 \approx 2.0 - 2.6$	• If we take walking frequency $F_{walk} \approx 2.3$ which is a step mid-way between 2.0 – 2.6 • Fundamental frequency of LoCaDe-Exo = 14.2 Hz • Separation factor = $S = F_{device}/F_{walk} \approx 6.2$
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